POST-FIRE RESPONSE OF FOUR PLANT COMMUNITIES IN SOUTH-CENTRAL NEW YORK STATE[1]

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Abstract. Four plant communities—northern hardwoods, oak woods, goldenrod (*Solidago* sp.)-poverty grass (*Danthonia spicata*) fields, and little bluestem (*Andropogon scoparius*) fields—were studied in seventeen areas on west to southeast slopes which had been burned by wildfires in 1962, 1963, or 1964. Stability of oak woods after fire was greater than that of northern hardwoods as shown by the lower percentage of dead and dying trees (16% vs. 35%), higher percentage of saplings sprouting (87% vs. 43%), higher average number of sprouts per sapling (4.4 vs. 2.5), and higher percentage of ground-layer species remaining unchanged or increasing in frequency (87% vs. 67%). In goldenrod-poverty grass fields, fire caused goldenrods to increase and povery grass to decrease. Average weight of dicots was 24% more and that of monocots was 50% less in burns than in unburned portions. These fields appeared to revert to poverty grass dominance after several years without fire. In little bluestem fields, red top (*Agrostis alba*) and little bluestem increased after fire. Average weight of monocots was 32% more than that of dicots was 35% less in burns than in unburned portions. Wildfires top-killed 63–93% of saplings invading fields; 77–90% of saplings in burns produced sprouts.

INTRODUCTION

The purpose of this study was to measure vegetative response of different plant communities to accidental burning. Seventeen accidentally burned areas in south-central New York, representing four plant communities, were selected for study (Fig. 1). They included three northern hardwoods stands (areas 1–3), eight oak woods (areas 4–11, and six fields (areas 12–17).

In spite of a well-established protection system, wildfires have continued to be an important influence on plant communities in New York State. In 1962 and 1963 there were 1,532 and 1,429 fires, which burned 19,549 and 12,405 acres, respectively (New York State Conservation Department 1962, 1963). New York State Conservation Department records reveal that 73% of all fires in the study zone (Fig. 1) from 1946 to 1963 occurred in March, April, and May, aided by warm, dry periods (Schroeder 1950), brisk winds, and dead, dry vegetation.

Soils of the study areas were acid glacial tills of the Bath-Mardin-Lordstown Association (Cline 1955). European agricultural development reached its peak about 1880 and then rapidly declined (Vaughn 1929), leaving over 230,000 acres of fields in south-central New York to gradually change to forest by natural succession or artificial reforestation.

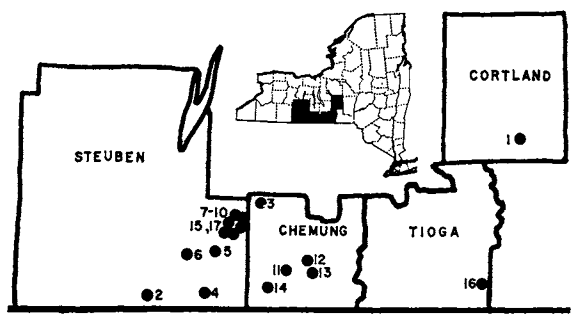

FIG. 1. Map of south-central New York showing locations of the 17 study areas.

I gratefully acknowledge the assistance of L. S. Hamilton in guiding this study and H. A. Regier in providing statistical advice. L. S. Hamilton, R. W. Bailey, J. D. Gill, and K. T. Rinell kindly reviewed the manuscript.

METHOD AND PROCEDURE

Seventeen accidentally burned areas were selected for study. Fires had occurred between April 6 and May 2, 1962, 1963, or 1964, except in area 17, which had burned in early September 1964. All were studied 10–26 months after burning. Study areas were located on 2–50% slopes facing west to southeast and were at elevations between 900 and 1,700 ft (274–518 m). Each burn had an unburned area adjacent to it similar in topography, vegetational composition, and history of land use. The four plant communities studied were (1) northern hardwoods, (2) oak

[1] A project of the New York State Agricultural Experiment Station at Cornell University, supported by McIntire-Stennis forestry research funds of the U. S. Department of Agriculture and conducted with the cooperation of the New York State Conservation Department. (Manuscript received January 22, 1970; accepted June 15, 1970.)

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woods, (3) goldenrod (*Solidago* sp.)-poverty grass (*Danthonia spicata*) fields, and (4) little bluestem (*Andropogon scoparius*) fields.

Each burned and adjacent unburned area contained at least 2.8 ha, and three plots 33 m square were randomly located within each member of a pair. Seven random points were located in a plot. Frequency and density of ground-layer species were recorded within a square-meter quadrat, a corner of which was on a random point. Each area contained 21 sampling points or quadrats. Within a quadrat each species with plants less than 1.4 m tall was recorded as having 1–5, 6–25, or more than 25 stems. At each point soil pH was determined, and trees and saplings along with any basal sprouts were sampled by the quarter method (Cottam and Curtis 1956). In forest study areas litter samples were collected at each point; in fields separate clippings were made of living monocots, living forbs and woody perennials 1.3 cm or less in diameter at the base, and dead plant material (natural mulch). Loss-on-ignition properties of soils are discussed in Swan (1968).

Analyses of variance for the following data were carried out at the Cornell University Computing Center: pH at the soil surface and 6-cm depth; litter weights for 11 forested study areas; for six fields, weights of clippings of natural mulch, living monocots, living dicots, and the combined weights of dicots and monocots. The data were separated into two groups, forests and fields, and F tests were carried out by use of the analysis of variance for the randomized complete block design (Steel and Torrie 1960). The data were then divided into four groups on the basis of vegetational differences, and F tests were made. The null hypothesis of no difference between burned and unburned areas of a plant community was accepted if the test's result was less than F (5%) values in Steel and Torrie (1960).

The chi-square test was used to determine response of a ground-layer species to burning. Quadrat data were tested on species that had an average frequency (AF) of 15 or more in either half of two or more study areas. For species with high AF in only one study area, the test was made when the expected number of occupied quadrats was five or more (Greig-Smith 1964). Two null hypotheses were tested: (1) that a species occurred in as many quadrats in burned as in unburned areas; and (2) in those burned areas where a species occurred, abundance (number of occupied quadrats in each density class) was the same as in unburned stands. On the basis of tests of these two hypotheses, species in each community were divided into increasers, decreasers, neutrals, invaders, and retreaters (Vogl 1964*a*). Retreaters were species present in the unburned sample and absent in the burned sample. Invaders were species found only in the burned sample. Neutral species showed no significant difference in AF between condition pairs.

Voucher specimens of most of the plant species sampled and photographs of the study areas are on record at the Department of Conservation, Cornell University. Plant nomenclature follows Clausen (1949) and Fernald (1950).

Results

Soil pH and plant production

Soil surface pH values in burned areas were significantly higher than those in unburned areas of both forests and fields (Table 1). In two northern hardwoods stands 1 year after fire, burned soil surface pH averaged 1.6 and 0.6 units higher, and the range of pH values was 4.3 to 7.2, which was nearly twice the unburned range (4.0–5.6). In a third northern hardwoods stand, the average difference in soil surface pH between areas was only 0.2 unit 2 years after fire. Average soil surface pH of the burned oak woods was significantly higher by 0.6 unit than that of unburned portions. The average range was 4.3 to 7.2 in

TABLE 1. Average values and results of analyses of variance of six variables measured in 17 adjacent unburned (U) and burned (B) areas, representing four plant communities in south-central New York (all weights expressed in g/0.25 m²)

Item	All forests		All fields		Northern hardwoods		Oak woods		*Solidago* fields		*Andropogon* fields	
	U	B	U	B	U	B	U	B	d	B	U	B
pH, soil surface	5.0**	5.6	5.0*	5.2	4.8	5.6	5.0*	5.6	5.0	5.3	4.8	5.1
pH, 6-cm depth	4.5	4.7	4.8	4.8	4.5	5.0	4.5	4.6	4.8	4.9	4.7	4.8
Weight, litter or natural mulch	71.4**	48.7	71.0**	28.9	65.4	36.4	73.7**	53.2	69.0*	33.5	75.2	19.7
Weight, monocots	—	—	—	—	—	—	—	—	19.6*	9.7	30.2	39.9
Weight, dicots	—	—	—	—	—	—	—	—	40.1	49.9	24.8	16.0
Weight, living plants	—	—	58.1	58.4	—	—	—	—	59.7	59.6	55.0	55.9

*Significant difference between unburned and burned areas
**Highly significant difference between unburned and burned areas

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burned areas compared with an unburned range of 3.8 to 6.8. Soil surface pH of six burned fields averaged 0.2 unit greater than that of unburned portions. Average range of burned values was 4.8 to 6.0, and that of unburned values was 4.6 to 5.4. In all study areas except one northern hardwoods stand, the difference was slight between burned and unburned pairs in soil pH at 6-cm depth.

Litter weights over all burned forest areas were less than those in unburned portions to a highly significant degree (Table 1). This difference was highly significant over the eight oak stands by the F test and in one northern hardwoods stand by the t test. Weights of natural mulch in fields unburned for three to more than 18 years were greater than those in fields unburned for 1 year to a highly significant degree (Table 1). The F test showed significant differences between conditions in four *Solidago* fields but not in two *Andropogon* fields. After 1 year of no fire, average mulch accumulation in burned portions of *Solidago* fields was 48% of the unburned average and in burned portions of *Andropogon* fields was 26% of the unburned average.

Agronomic production of living monocots, primarily grasses, in the two field communities differed in response to burning. In *Solidago* fields monocots had an average weight per unit area 50% less in burned than in unburned portions (Table 1). In *Andropogon* fields monocots had a 32% greater weight in burned portions. Agronomic production of woody and herbaceous dicots in the two field communities also differed. Burned portions of *Solidago* fields had a 24% greater average weight of dicots than unburned portions. Burned *Andropogon* fields had a 35% smaller average weight of dicots than unburned areas. Agronomic production of all living plants in *Solidago* fields was 7% greater than that of *Andropogon* fields (2,386 and 2,218 kg/ha, respectively, from averages of both condition pairs in each community).

TABLE 2. Distribution of ground-layer vegetation in behavioral categories in 17 burned areas, representing four plant communities, expressed as the percentage of the total number of species in a community in each category

Category	Northern hardwoods	Oak woods	*Solidago* fields	*Andropogon* fields
Number of species tested	24	52	43	27
Neutrals (%)	38	54	46	63
Increasers (%)	29	33	26	7
Decreasers (%)	12	13	28	26
Invaders (NH) or retreaters (%)	21	0	0	4

TABLE 3. Average frequencies (%) of ground-layer species in one black oak-red maple and three northern hardwood stands with significant differences in frequency between burned and unburned portions

Species	Average frequency (%)	
	Unburned	Burned
Increasers		
Aster divaricatus	19	33
Carex sp.	6	24
Eupatorium rugosum	10	48
Galium triflorum	2	29
Prunus pensylvanica	3	19
Rubus sp. (blackberry)	8	29
Viola latiuscula, V. rostrata	16	48
Invaders[a]		
Amphicarpa bracteata	0	29
Circaea quadrisulcata	0	71
Geranium maculatum	0	48
Impatiens capensis	0	52
Solidago graminifolia	0	48
Decreasers		
Acer pensylvanicum	25	5
A. saccharum	29	5
Fraxinus americana	43	20

[a]Includes 14 other species with average frequency of 14 or more in the burned portion of one study area.

Species abundance

The percentage distribution of ground-layer species of each community was determined for each of the five behavioral categories (neutral, increaser, decreaser, invader, and retreater) (Table 2). Some species differed significantly in average frequency or abundance between burned and unburned conditions (Tables 3–6). Lists of neutral species are given in Swan (1966).

Northern hardwoods stands had the smallest percentage of neutrals and the largest percentage of invaders of the four communities. All three decreasers in northern hardwoods stands (Table 3) were seedlings of tree species: sugar maple (*Acer saccharum*), white ash (*Fraxinus americana*), and striped maple (*Acer pensylvanicum*). Two of the seven increasers were pin cherry (*Prunus pensylvanica*) and *Rubus* sp., which have often been described as fire followers. Invaders included species of surrounding fields with windborne fruits.

Oak woods had the highest percentages of increasers and neutrals of the four plant communities, and there were no invaders or retreaters (Table 4). This was the only community with two decreasers significantly more abundant in burns (higher densities in occupied quadrats): bracken fern (*Pteridium aquilinum*) and sassafras (*Sassafras albidum*). *Carex* sp. was the only

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TABLE 4. Average frequencies (%) of ground-layer species with significant differences in frequency or abundance between burned and unburned portions of eight oak woods

Species	Average frequency (%)	
	Unburned	Burned
Increasers		
Apocynum androsaemifolium	14	39
Carex pensylvanica	46	63
Carya sp.	7	18
Ceanothus americanus	17	33
Comandra umbellata	15	26
Corylus cornuta	8	25
Helianthus divaricatus	20	46
Maianthemum canadense	13	27
Melampyrum lineare[a]	1	11
Panicum dichotomum	4	22
P. latifolium	9	19
Poa compressa	6	17
Vicia caroliniana	15	31
Neutrals more abundant in burn		
Quercus sp. (black oak)	27	26
Potentilla simplex	10	17
Rhododendron nudiflorum	18	15
Waldsteinia fragarioides	36	43
Decreasers more abundant in burn		
Pteridium aquilinum	69	56
Sassafras albidum	62	43
Decreasers		
Acer rubrum	32	14
Diervilla lonicera	15	5
Fraxinus americana	71	29
Quercus alba	16	10
Vaccinium vacillans	47	36

[a]Chi-square test based on frequencies less than 15% from five study areas.

TABLE 5. Average frequencies (%) of ground-layer species with significant differences in frequency or abundance between burned and unburned portions of four *Solidago* fields

Species	Average frequency (%)	
	Unburned	Burned
Increasers		
Agrostis alba	25	39
Ambrosia artemisiifolia	2	33
Crataegus sp.	29	50
Galium boreale	2	33
Lithospermum arvense	24	57
Poa compressa	56	81
Populus tremuloides	2	31
Solidago juncea	90	99
S. nemoralis	34	50
Neutrals more abundant in burn		
Potentilla simplex	93	86
Rumex acetosella	33	44
Neutral less abundant in burn		
Trifolium pratense	25	31
Decreasers		
Achillea millefolium	76	36
Aster novae-angliae	29	5
A. umbellatus	62	10
Chrysanthemum leucanthemum	67	48
Cornus racemosa	62	44
Danthonia spicata	87	61
Daucus carota	38	21
Panicum lanuginosum	44	13
Rubus flagellaris	31	23
Rudbeckia hirta	29	2
Solidago rugosa	31	7

increaser of oak stands in common with those of northern hardwoods.

Solidago fields contained a lower percentage of neutral species than *Andropogon* fields, but a much greater percentage of increasers (Table 2). Flowering stalks of *Solidago juncea* were more abundant in burned portions of *Solidago* fields, averaging 7.2 per quadrat compared with 1.6 in unburned portions. Although this difference was large, it narrowly missed significance by the F test. Poverty grass (*Danthonia spicata*) was the most conspicuous decreaser of the monocots as shown by the large difference in average frequency of 26% between burned and unburned fields (Table 5). Also frequent patches of bare soil in burned fields contrasted with continuous mats of poverty grass in undisturbed fields.

The main feature of burned *Andropogon* fields compared with unburned fields was the significant increase in average frequency of the dominant little bluestem (*Andropogon scoparius*) (Table 6). Red maple (*Acer rubrum*), the principal tree invader of undisturbed fields, was an important decreaser. Canada blue grass (*Poa compressa*) and *Solidago juncea* were neutrals here, but increasers in *Solidago* fields.

Effects of fire on trees and saplings

Burned northern hardwoods stands had 35% dead or badly injured trees compared to 1.3% dead trees in unburned areas. Sixty per cent of hemlock (*Tsuga canadensis*) trees sampled in burns were dead or badly injured. Comparable values for sugar maple and beech (*Fagus grandifolia*) were 44% and 38% killed. There were twice as many dead or badly injured trees in burned oak woods as unburned areas, 16% and 8% respectively. Only small percentages of trees produced sprouts in both burned northern hardwoods and oak stands.

Saplings in burned northern hardwoods showed low resistance to fire compared to those of oak woods (Table 7). Burned northern hardwoods stands averaged 1,213 saplings per acre (491/ha), of which 49% produced sprouts after fire. Burned oak stands averaged 1,078 saplings per acre, and 87% sprouted after fire. There were also more sprouts per sapling in burned oak stands than in burned northern hardwoods (4.4 vs. 2.5). In all three burned northern hardwoods stands, saplings

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TABLE 6. Average frequencies (%) of ground-layer species with significant differences in frequency or abundance between burned and unburned portions of two *Andropogon* fields

Species	Average frequency (%)	
	Unburned	Burned
Increasers		
Agrostis alba	17	38
Andropogon scoparius	81	100
Neutrals less abundant in burn		
Danthonia spicata	43	52
Rubus allegheniensis, R. flagellaris	74	57
Decreasers		
Acer rubrum	62	29
Achillea millefolium	86	48
Anthoxanthum odoratum	33	12
Aster laevis	60	36
Pycnanthemum tenuifolium	81	33
Retreater		
Sericocarpus asteroides	33	0

of the dominant deciduous tree species, beech or sugar maple, produced the majority of sprouts or root suckers, although large numbers of saplings were completely killed without resprouting. Hemlock saplings suffered 93% mortality and did not resprout. In all oak stands but one, saplings of black (*Quercus velutina, Q. borealis*), white (*Q. alba*), and chestnut (*Q. prinus*) oaks were abundant. All three species produced sprouts in proportion to their preburn sapling importance values (Curtis and McIntosh 1951, Whitford and Salamun 1954).

More sprouts were produced from a single root system as sapling size increased in all species except hop hornbeam (*Ostrya virginiana*). Sapling size classes were 2.5 cm or less dbh, 2.6–5 cm, and 5.1–10 cm. Chestnut oak and red maple showed the greatest increases from the smallest to the largest size class: 4.9 to 12 and 4.3 to 10.7 sprouts per dead sapling, respectively.

In one burned northern hardwoods stand, 57% of the sprout groups, each growing from a single root system, were browsed, and in one burned oak stand, 62%. Heavy browsing was noted in other burns but was not measured.

Effects of fire on scattered saplings were studied in two fields. Saplings were stocked at densities of 815 and 415 stems per acre. Red maple and hawthorn (*Crataegus* sp.) were the principal invaders (Table 7). Although red maple sprouted vigorously in both burned fields and forests, 27%, of its saplings died without sprouting in fields, whereas none behaved this way in burned forests. In contrast, a higher percentage of hawthorn saplings were not top-killed in fields than in forests, and these survivors produced 6.8 sprouts per stem. White pine (*Pinus strobus*) saplings were greatly reduced in number by fire. Oaks were minor invaders of these fields, contributing 2.6% of all saplings.

TABLE 7. A comparison of sprouting abilities following fire of woody species with stems 10 cm or less dbh in 11 forested areas and two fields (Asterisks mark species common in northern hardwoods stands but rarely seen in oak woods. Those not marked were common oak woods species.)

Species	Total number of stems sampled	Percentage of total sample dead without sprouts	Percentage of stems not top-killed	Number of sprouts per top-killed stem
Forests				
Cornus florida	10	0	50[a]	7.2
**Acer pensylvanicum*	8	25	0	6.4
Quercus prinus	65	0	11	5.9
Crataegus sp.	11	0	9	5.8
Quercus alba	73	11	18[b]	5.5
Castanea dentata	16	0	0	5.0
Acer rubrum	65	0	2	4.8
**Ostrya virginiana*	71	35	3	4.4
Corylus cornuta	12	8	0	4.1
Quercus sp. (black oak)	149	7	17[c]	3.6
Carya sp.	73	4	28[d]	3.5
**Acer saccharum*	48	44	10	3.2
**Fagus grandifolia*	55	53	5	2.5
**Fraxinus americana*	22	27	9	1.8
**Tsuga canadensis*	43	93	7	0
Fields				
Quercus alba	5	0	40	6.0
Cornus racemosa	8	12	0	5.1
Crataegus sp.	123	4	31[e]	4.7
Acer rubrum	232	27	4	4.5
Prunus serotina	8	12	0	4.4
Quercus sp. (black oak)	7	0	29	4.4
Amelanchier sp.	18	0	11	4.3
Populus tremuloides	37	8	5	2.6
Corylus americana	7	0	0	1.0
Pinus strobus	6	83	17	0

[a]7.2 sprouts per surviving sapling.
[b]1.0 sprout per surviving sapling.
[c]2.1 sprouts per surviving sapling.
[d]0.3 sprout per surviving sapling.
[e]6.8 sprouts per surviving sapling.

A burned and unburned pair of aspen (*Populus tremuloides*) stands was studied in two other fields. In one field 99% of the aspen stems were less than 5 cm dbh. Stand density was 15,500 stems per acre. In the burn, all stems sampled were killed, and there were 13.8 root suckers per square meter. The adjacent unburned stand had 1.0 seedling or sucker per square meter. In the other field 79% of the aspen stems were less than 5 cm dbh, and aspen density was 5,866 stems per acre. Sixty-six per cent of all stems in the burn were dead, and 3.8 root suckers per square meter were present. Sucker density in the unburned stand was $0.3/m^2$. These results should be interpreted with caution because of the small number of stands sampled,

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especially in the northern hardwoods and *Andropogon* communities.

Discussion

Northern hardwoods

Because of its root-suckering ability, beech remained the dominant species in one study area that was burned by a light ground fire. Ground fires could create favorable conditions for production of beech suckers by reduction of litter and humus layers or by injury to parent trees. Many of the largest beech trees were hollow, perhaps because of fires more than 40 years before the current burn. The presence of a few large, well-decayed stumps indicated that the area was once logged. Although all hemlock saplings were killed in the most recent fire, some trees survived and might restock the burned area. Because this light fire consumed only small amounts of the litter in this beech stand, no change was noted in the ground-layer vegetation, which consisted of only 15 species. Only a small amount of litter was consumed probably because of the speed of the fire and moistness of the organic layer.

Sugar maple and hop hornbeam were favored in the two burned sugar maple stands because their saplings produced 34% and 46% of all sprouts, respectively. Because of this hot fire, large amounts of litter and humus were consumed, which resulted in areas of high pH and great changes in the ground-layer vegetation. In many places the depth of the organic horizon had been reduced 7–10 cm, especially next to trees, on which fragments of the former surface adhered. Burned areas of these two stands averaged 62 species, whereas unburned areas averaged 40. Pin cherry, an increaser (Table 3), had stems more than 1.4 m tall only 15 months after fire. In addition to five species classed as invaders, 14 others, including trembling aspen and bitternut hickory (*Carya cordiformis*), had an average frequency of 14 or more in the burned area of one of the two stands and were absent from the unburned area.

Wilm (1936) studied the vegetation of logged areas after severe slash fires in Schuyler County, New York, adjacent to Steuben and Chemung Counties. He found that aspen and cherry were often the first tree species to occupy such burns.

Oak woods

When compared with northern hardwoods stands, oak woods showed greater fire resistance by their lower percentage of dead and dying trees (16% vs. 35%), higher percentage of saplings sprouting (87% vs. 43%), and higher average number of sprouts per sapling (4.4 vs. 2.5). There were at least 1,370 more sprouts per acre in the burned oak woods than in the burned northern hardwoods stands. In the oak woods burned two or more times before this study, only one of several groups of sprouts from a single root system or basal plate was sampled.

In the oak woods study areas, counts of annual growth rings of saplings of sprout or sucker origin showed that in the 20 years prior to this study 14 fires had occurred in burned sections and five in unburned areas. These figures included the most recent fires. This difference was significant by the chi-square test and indicated that past fires tended to start in the same areas and to follow the same paths, aided by dryness of south slopes and flammability of dried vegetation. Similar reasoning could explain the high frequency of fires in field study areas, nine fires in burns and four in unburned areas.

Three species of oaks (white, chestnut, and black) produced large numbers of sprouts (3.6–5.9) per top-killed sapling because of their inherent ability to sprout and large food reserves still present before bud break (Hodgkins 1958). Oak saplings also had high survival rates (11–18%). These characteristics have insured the perpetuation of oak forests in the presence of frequent ground fires. A correlation between disturbance and oak regeneration has been reported in West Virginia (Carvell and Tyron 1961), southern Wisconsin (Rogers 1959), southern Connecticut (Olsen 1965), and Rhode Island (Brown 1960).

Repeated fires in these oak woods significantly reduced seedlings and saplings of red maple and other northern hardwoods species (Tables 4 and 7). Four of the seven decreasers were tree species, including red maple, white oak, white ash, and sassafras. Since all top-killed red maple saplings sprouted, it took only several years without fire for competition from red maple to increase rapidly. After 20 years of growth, red maple stems exceeded 10 cm dbh and had thick enough bark to survive ground fires.

A fire-resistant community of trees, shrubs, and herbs has been selected by a long history of recurring fires in the oak stands, since only 13% of the ground-layer species were decreasers. Of the six species of ericaceous shrubs present, *Vaccinium vacillans* decreased, while five were neutrals (*Gaultheria procumbens, Gaylussacia baccata, Rhododendron nudiflorum, Vaccinium angustifolium,* and *V. stamineum*). Bracken fern and bush honeysuckle (*Diervilla lonicera*) were the other two decreasers. Although bracken fern decreased significantly in average frequency, it was more

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abundant in burns to a highly significant degree. Existing clumps must have contracted and become more dense. In a study of 27 burned and unburned pairs of bracken grasslands in northern Wisconsin, Vogl (1964*b*) also found that bracken fern and bush honeysuckle were decreasers.

One black oak stand was successionally intermediate between the oak woods and northern hardwoods. Red maple was second to black oak with a tree importance value of 76. This stand had not burned for more than 20 years; abundant red maple saplings and seedlings were present, but those of black oak were absent. Therefore the ground fire under study favored sprout production of more mesic species, such as red maple, hop hornbeam, and flowering dogwood (*Cornus florida*). The dominant position of red maple in this stand and its potential importance in the other oak stands in the absence of fire indicate its role as a bridge in the transformation of oak woods in the study zone to northern hardwoods (Larson 1953). Taylor (1928) stated that fire was the most important factor preventing the oak-hickory forests of Allegheny State Park, in southwestern New York, from changing to northern hardwoods. He assigned to red maple and white pine the role of successionally linking the oak-hickory and northern hardwoods forests.

The predominance of spring fires over autumn fires in the study zone for the past 30 years may reflect a long-prevailing regime of spring fires that has existed at least since European settlement. This situation could explain the dominance of black oak in the sprout, sapling, and tree layers. Korstian (1927) stated that white oak acorns germinated in the autumn, whereas those of black oak germinated in the spring. He listed the oak species in order of resistance of their acorns to the high temperatures of ground fires: red oak, chestnut oak, black oak (*Q. velutina*), scarlet oak (*Q. coccinea*), and white oak. Spring fires would kill white oak seedlings and give those of black oak a competitive advantage since black oak acorns were more heat resistant and would germinate after the fires.

Wiegand and Eames (1925: 441) stated that oak-chestnut forest was more abundant south and southwest of the Cayuga Lake basin than to the east where stands of northern hardwoods forest became larger and more numerous. They concluded that "The contrast between these two types of forest is altogether striking, but the factor or factors determining their occurrence are apparently not known and the problem is an exceedingly interesting one." An interaction of climate, fire, and topography over many centuries could have caused this striking contrast in vegetation.

Many authors have described the important role that recurring fires have played in increasing the area of xeric communities such as prairies and pine and oak forests (Gleason 1913, Harper 1913, Sauer 1950, Curtis 1959). From 1946 to 1963 more fires occurred in the southwestern part of the study zone. Of an annual average of 1,630 fires, distribution by county was Steuben, 1,032; Chemung, 474; Tioga, 94; and Cortland, 30 (New York State Conservation Department 1946–1963). Probably this pattern has existed for centuries because of an interaction of drier climate, Indian and lightning fires, and fewer natural firebreaks southwest of the Cayuga Lake Basin (Day 1953). Larger areas of northern hardwoods developed to the east of the basin as the result of a cooler, moister climate and more natural firebreaks, such as the Finger Lakes and deep north-south valleys.

Effects of fire on fields

In burned *Solidago* fields, the significant decrease in production of monocots (mostly poverty grass and Canada blue grass) was balanced by the increase in weight of dicots (Table 1). Greater numbers of flowering stalks of *Solidago juncea, S. bicolor,* and *S. nemoralis* contributed most to the 24% greater average weight of dicots on burns. In Illinois Rice (1932) found that burning caused forbs to increase over grasses on a railroad right-of-way that contained predominantly non-fire-resistant blue grasses, which begin growth earlier in spring than native prairie grasses. After fire, *S. juncea* increased in cover by 16% in New Jersey (Jordan 1965) and in numbers of stems at Ithaca, New York (Baker 1968). Results of this study agree with those of Beckwith (1954), who concluded that infrequently burned fields in Michigan had about the same vegetation as unburned ones, but that regular burning of fields would maintain indefinitely a community composed of blue grasses, goldenrods, and asters. With the passage of fireless years, flowering of goldenrods should progressively decrease, and poverty grass should regain dominance.

Unburned *Solidago* fields were covered by a shallow mat of twisted blades of poverty grass with scattered basal rosettes of goldenrods. From a quantitative sample of 24 unburned fields in central New York, Abel (1941) found that poverty grass invaded soon after abandonment and formed a thick mat, which remained on fields until trees increased enough to shade out that community. Russell (1955) reported little change over 30 years in six unburned fields in eastern New York, which were dominated by poverty grass and goldenrods. In this vegetation type, therefore, fire

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temporarily increased forb dominance and inhibited invasion of some woody perennials.

Average mulch accumulation in unburned *Solidago* fields was slightly less than that of *Andropogon* fields (2,760 vs. 3,008 kg/ha), and after fire natural mulch should accumulate to preburn amounts in 4 years. The small difference between condition pairs in annual agronomic production, which averaged 1.6 tons per acre, indicated an equally efficient recycling of nutrients in burned and unburned areas.

In burned *Andropogon* fields, the average weight of forbs decreased 35%, but grasses, principally little bluestem, increased 32% in average weight. All species of goldenrods remained neutrals. The inability of *S. juncea* to increase after fire when in competition with little bluestem compared with its increase in the burned poverty grass community also confirmed the vigorous post-fire response of little blustem.

In the study fields, fire completely killed 10–23% of invading saplings. Surviving saplings, mostly red maple, hawthorn, aspen, and shadbush (*Amelanchier* sp.), produced dense clumps of sprouts, much shade, and competition for herbaceous plants within 2 years. Beckwith (1954) also reported that hawthorn was common in grazed or burned fields of Michigan and was resistant to fire due to its abundant sprout production and high survival rate. A single fire seemed to aid the spread of trembling aspen clones into fields (Vogl 1969), since it had an average frequency of 31 in burned *Solidago* fields and 2.4 in unburned ones. In fields of this study and of Abel's (1941), oaks and white pine were uncommon, although white pine is a major invader of old fields in the study zone where seed sources are available.

Rapid successional change was measured by Bump (1950), who found that from 1927 to 1947 90% of the open lands in the Connecticut Hill Game Management Area had passed to thickets of trees and shrubs in the absence of fire. If the land management objective was to maintain open fields in the study zone, prescribed fires at 3- to 5-year-intervals would be needed to reduce the density of woody plants and keep sprouts from shading herbs.

Adjacent to some of the oak woods study areas were isolated pockets of oak scrub, which had also burned frequently in the past and had not been used intensively for grazing. Species grew there which are typical of fire-maintained oak openings and prairies to the west, such as *Andropogon gerardi, Asclepias tuberosa, Desmodium canadense, D. paniculatum, Lespedeza capitata, L. intermedia, Lupinus perennis,* and *Sorghastrum nutans*. With such a reservoir of fire-resistant species in the study zone, a program of prescribed burning could be confidently undertaken for maintenance of oak woods and fields as natural areas, wildlife management areas, and soil-restoration projects. Both old fields and pole-stage forest stands could be made more productive in wildlife by a program of introduction of local ecotypes of fire-resistant species and prescribed burning.

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